Tactix: Lifted-Proof Repair and Correctness-Gate Checkpoint

Report to Andrew

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Outcome

The recent native-LRAT hint audit found three producer-level defects. Instantiate could cite a root unit already assumed by its lemma cube; Strengthen and BVE could cite the same root unit twice. Those were repaired with focused causal regressions. A fourth issue in Sweep's general binary ladder reused a subsolver chain after lifting its lemma by a target literal. The chain is now replayed under the *complete* negated published lemma while the original environment still exists. Already-satisfied hints are removed, and a chain that cannot close is declined before any kernel mutation. This is a general proof condition, not an instance-specific workaround. One proposed BVE definition-unit change was disproved by the exact counterexample and reverted.

Evidence and limits

On four unchanged original-CNF handoffs, a pinned Debug public-API run (SLURM job 1722204, four CPUs, 32 GiB) emits proofs with zero strict-LRAT audit failures; CakeLPR independently verifies all four. The BVE case includes 41 audited RAT lines. Fresh Release focused tests pass 7/7, and ASan+UBSan focused rule tests pass 4/4. Exact commands, hashes, and checker logs are in the AM421 run; the analysis is in the AM421 technical report. These tests do not replace full current-source fuzz or the combined N1/N2 gate.

I independently classified the three AM400 DRAT strict-audit flags. In each, every proof addition before the first root contradiction is RUP or RAT; the audit flags only a later trailing empty addition after deletions have removed that contradiction from its reconstructed population. The valid refuting prefix remains sufficient, and drat-trim's backward and forward checks plus CakeLPR verify the proofs. This is an audit-harness state error after terminal refutation, not a new solver inference defect. The historical campaign is unchanged; any shared audit repair needs positive and negative self-tests. The AM400 Probe/Lucky owner has terminal affected-scope PASS, not universal N1/N2 acceptance. AM401, AM412, and AM414 owners remain active on their own immutable pins.

Readiness and next actions

The canonical no-option DSR-to-LSR-to-Trestle H2 rejection remains open; no checker route was relaxed. General live pseudo-Boolean arithmetic, redundancy and dominance remain a qualified winner-proof-calculus gap, while Tactix's PBP production is terminal startup-only. The earlier architecture assessment still holds: online learning can prototype ordered and budgeted root passes, but production learning needs a shared default-exact slot/dose controller and owned safe-point switching for branching and deletion sub-strategies. Neither the combined N1–N4 gates nor the performance gates are satisfied. Resolve H2 and remaining owner handoffs, run current-source fuzz and full proofs/models, then remeasure the gates before the requested Tier-D comparison (5,000 seconds, four CPUs, 32 GiB, hwulab). No

Tier-D job has been submitted. The living calculus and implementation notes meet their line budgets exactly; documentation compaction was not requested.